Kudumula Gopi Charan Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.69
Sri Chaitanya	Class XII (Telangana State Board of Intermediate Education (TSBIE))	2023	98.50%
Sri Chaitanya	Class X (Board of Secondary Education, Telangana (BSET))	2021	10.00%

WORK EXPERIENCE

Oracle -- Software Engineering Intern

Summer 2026

- Added Oracle Database support to **Graphiti**, an open-source temporal knowledge graph framework, enabling Oracle as a supported backend for graph storage and retrieval.
- Implemented optimal integration, including schema design, driver compatibility, and query optimization for temporal graph operations.
- Conducted latency checks and performance benchmarking to validate low-latency reads/writes and identify bottlenecks at scale.
- Collaborated with the open-source community to upstream changes and ensure compatibility with Graphiti's existing architecture.

SCHOLASTIC ACHIEVEMENTS

- Achieved **Expert** Level on Codeforces
- Achieved **7th** place in FedEx PS, Inter IIT Tech Meet 13.0
- Achieved **4th** place in Motive Hackathon

POSITIONS OF RESPONSIBILITY

Core at Lambda - The Software Development Club

- Contributed to the backend development of the college dashboard using **Golang** and Integrated the backend with a PostgreSQL database.
- Worked with a team using **Git**, followed modular coding practices, and participated in code reviews.

Core at Infero - The Algorithmic and Logical Reasoning Club

- Conducted **DSA sessions**, set internal contests, and mentored juniors in advanced problem-solving

SKILLS

- Languages : C, C++, Javascript, Golang, Python, SQL, HTML, CSS, R
- Frameworks : React.js, Go Gin, Express.js, Node.js, Numpy, Matplotlib, FastAPI.
- Tools : Git, Github, Latex, VS-code, Ollama

- AI Frameworks : Pandas, PyTorch, TensorFlow, Scikit, Hugging Face, LangChain, LlamaIndex, CrewAI

RELEVANT COURSES

- **AI**: Foundations of Machine Learning, Deep Learning*, Multi Agent Systems*, NLP, Practical Challenges in Image Analysis, Business Analytics using Python, Convex Optimization, Predictive Modelling.

- **Computer Science**: Intro to Programming, Data Structures and Algorithms(Theory and Lab), Algorithms, Theory of Computation, DBMS-1, Compilers-1, Intro to AI, Operating Systems, Software Engineering.

- **Mathematics**: Combinatorics and Graph Theory, Linear Algebra, Convex Optimization, Probability, Applied Statistics, Introduction to Number Theory, Group Theory and Rings, Calculus(1 & 2), Vector Calculus, Multivariable Calculus, Ordinary Differential Equations, Metric Spaces, Transform Techniques, Real Analysis, Spectral Graph Theory.

PROJECTS

Smartphone Eye Image Quality Enhancement · Deep Learning / Medical Imaging Project · GitHub

- Built a transformer-based image restoration pipeline (Restormer) to transform low-quality smartphone eye images into slit-lamp clinical quality, using MDTA and GDFN for efficient 512×512 processing.
- Generated paired training data (~1,719 images) via Neural Style Transfer and implemented a lightweight U-Net segmentation module with a dedicated testing suite.

Mental Manipulation Detection Pipelines · NLP Research Project · GitHub

- Built dual NLP pipelines for detecting mental manipulation in dialogue on the MentalManip dataset (4,000 annotated movie dialogues).
- Developed an Intent-Aware Prompting (IAP) pipeline with async 3-step LLM orchestration and a unified client supporting OpenAI, Anthropic, and Groq; targeted reduced false negatives over single-pass baselines.
- Implemented a similarity-based pipeline using sentence embeddings with threshold tuning and emotion vector mapping for technique-label prediction.

3D Bin Packing Solver using Genetic Algorithms · Inter IIT Tech Meet 13.0 - FedEx Challenge, Group Project · Github

- Designed and implemented a 3D bin packing algorithm using genetic algorithms to optimize space utilization for package logistics, addressing constraints like orientation, volume, and container size.

- Achieved notable improvements in packing efficiency over greedy heuristics by evolving solutions through genetic operations such as mutation, crossover, and selection, as part of a real-world FedEx problem statement.